

AI Music Generation — Project Proposal

Emotion-Controllable Personalized BGM Generation — Continuous Emotion Control on MusicGen

Abstract

This project designs and implements an end-to-end AI music generation system that produces personalized background music (BGM) from a user's emotional intent. The system fine-tunes the pretrained MusicGen model rather than training from scratch, keeping it feasible within limited compute and time. Its core innovation is continuous emotion control, addressing a common weakness of current text-to-music (TTM) models—output biased toward emotional neutrality with weak controllability—so that abstract affective intent is mapped into stylistically coherent music.

1. Background & Motivation

Recent models such as **MusicGen**, MusicLM, and AudioLDM can already generate high-quality music from natural-language descriptions. However, **emotional controllability remains a clear weakness**: a recent benchmark (Almoclips, 2025) finds that current TTM systems are generally biased toward emotional neutrality and struggle to convey a specified emotion accurately and adjustably.

This is exactly what our topic—“turning emotion into bespoke BGM”—calls for. Rather than *spreading thin across emotion, style, and structure*, this project focuses on the emotion dimension in depth: letting users specify emotion in a continuous, quantifiable way and making the generated music faithfully reflect it.

2. Preliminary Literature Review

2.1 Audio Discretization & Coding (EnCodec / RVQ)

EnCodec is a convolutional auto-encoder that compresses continuous audio waveforms into discrete tokens via Residual Vector Quantization (RVQ): multiple codebooks encode the signal stage by stage, with each quantizer encoding the residual error of the previous one (8 codebooks in MusicGen). This turns audio into a text-like discrete sequence, enabling language-model-style modeling.

2.2 Autoregressive Generation (MusicGen)

MusicGen uses a single-stage Transformer to autoregressively model EnCodec tokens, predicting multiple codebooks in parallel via a delay/interleaving pattern. Its text conditioning is provided by a **T5 text encoder**, without relying on diffusion.

2.3 Conditioning & Evaluation (CLAP / MusicCaps)

CLAP maps text and audio into a shared semantic space via contrastive learning; here it is used mainly for evaluation (CLAP score, measuring text–music alignment) and automatic emotion labeling. MusicCaps is a music dataset with detailed text captions, often used as an evaluation benchmark.

2.4 Unified Framework (AudioCraft)

Meta's open-source AudioCraft provides unified implementations of MusicGen and EnCodec, serving as the engineering basis for our reproduction and extension.

3. Objectives & Innovations

Overall objective: Build an end-to-end, self-contained system that generates stylistically coherent, emotionally accurate personalized BGM from continuous emotion coordinates.

Innovations (deliberately focused, not scattered):

1. **Continuous emotion control (core).** Replace discrete labels (“happy/sad”) with a 2-D continuous Valence-Arousal coordinate for smooth, continuous emotion adjustment. We freeze the MusicGen backbone and train only a lightweight emotion-conditioning module (adapter), which is compute-friendly.
2. **Emotion-fidelity alignment (bonus).** Introduce a “generated emotion → target emotion” alignment loss during training, directly targeting the neutrality-bias problem.

4. Technical Approach

4.1 System Pipeline

Text description + Emotion coordinate (V-A) → conditioning-fusion module → MusicGen Transformer (frozen backbone) → EnCodec decoder → output audio

4.2 Training Strategy

- Freeze the pretrained backbone and train only the emotion-conditioning module (adapter or LoRA), greatly reducing compute and training time.
- Validate in stages: first verify controllable emotion conditioning on a small dataset, then scale up the data and tune.

4.3 Data

- Primarily music data with emotion annotations;
- Where annotations are scarce, use an emotion-recognition model / CLAP to auto-label unlabeled music with V-A values to expand the training set (data feasibility is a later focus).

5. Evaluation

Dimension	Metric / Method
Audio quality	FAD (Fréchet Audio Distance, lower is better)
Text–music alignment	CLAP score
Emotion fidelity	Emotion-recognition model measuring “generated vs. target emotion” consistency + small-scale subjective listening test

6. Timeline

Phase	Main Work
Phase 1	Literature review; reproduce a MusicGen inference baseline on AudioCraft
Phase 2	Data preparation & emotion labeling; implement the continuous emotion-conditioning module
Phase 3	Fine-tuning & hyperparameter tuning; validate emotion controllability
Phase 4	Evaluation & comparison experiments; system integration, demo, and report

7. Potential Extensions

To be attempted only after the core goals are met and if time permits.

- **Melodic-cue conditioning (primary extension).** Support user-provided melodic cues (humming / MIDI), generating BGM with a specified emotion while preserving the melodic contour—corresponding to the “melodic cues” in the topic. This requires an additional melody-alignment mechanism in the conditioning module and is more complex to implement, hence listed as an extension.
- **Multimodal emotion conditioning.** Infer emotion automatically from an image or short video, then generate matching BGM.

8. Feasibility

This project **fine-tunes an open-source pretrained model rather than training from scratch**; with a frozen backbone and only a lightweight conditioning module to train, both compute and timeline are manageable. The literature review and baseline reproduction have a clear path, the innovations are focused and can be validated in stages, and the risks are controllable.